

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F78AB

ACTUARIAL AND FINANCIAL MATHEMATICS B

Semester 2 2015/16

Duration: Two Hours

Total marks: 60

ANSWER ALL QUESTIONS

Electronic calculators approved by the University may be used.

1. Assume that time is measured in years. The annual force of interest $\delta(t)$ is a function of time and at any time $t \geq 0$ is given by

$$\delta(t) = 0.04 + 0.001t.$$

(a) Calculate the accumulated value of £1 accumulated from time $t = 0$ to time $t = 9$.

[3 marks]

The accumulated value of £1 accumulated from time $t = 0$ to time $t = 8$ is £1.42191.

(b) Calculate the accumulated value of £1 accumulated from time $t = 8$ to time $t = 9$.

[1 mark]

(c) Using your results from part (a) and part (b), calculate as annual effective rates:

i. The nine-year spot rate of interest from time $t = 0$ to time $t = 9$.

ii. $f_{8,1}$, where $f_{8,1}$ is the one-year forward rate of interest from time $t = 8$.

[2 marks]

[Total 6 Marks]

2. (a) Explain what is meant by a *forward contract*. Your answer should include reference to the terms *short forward position* and *long forward position*.

[3 marks]

(b) Three years ago, an investor entered into a long forward contract to buy a stock. The contract is due to mature in seven years' time from now.

The price of the stock was £82 three years ago and is now £54. The risk-free rate of interest can be assumed to be 5% per annum effective during the contract.

Calculate the value of the contract now if the stock will pay a single coupon of £6 in four years' time and this was known from the outset. No other coupon is paid. You should assume no arbitrage.

[5 marks]

[Total 8 Marks]

3. An engineering company has been awarded a contract to extend a tram line in a city and, once complete, to operate trams on it for a fifteen-year period. Construction of the tram extension is to start on 1 January 2017 and finish on 31 December 2018. The cost of the extension comprises a lump sum of £40 million payable on 1 January 2017 and £50 million per annum payable continuously at a constant rate during the calendar years 2017 and 2018.

From 1 January 2019, trams will begin running on the extended line. It is assumed that the net income will be received continuously at a constant rate of £17 million per annum for fifteen years.

(a) Calculate the net present value of the contract at 1 January 2017, at an effective rate of interest of 7% per annum.

[5 marks]

(b) Calculate the discounted payback period at an effective rate of interest of 7% per annum.

[5 marks]

(c) Without doing any further calculations, explain whether the discounted payback period would be greater than, less than or equal to that given in part (b) if the effective interest rate was reduced to 6% per annum.

[2 marks]

[Total 12 Marks]

4. An insurance company has liabilities of £10 million due in 12 years' time and £20 million due in 15 years' time. Its assets consist of two zero-coupon bonds, one maturing in 5 years' time and the other maturing in 25 years' time.

The current force of interest is 5% per annum.

(a) Calculate the amount that should be invested in each zero-coupon bond so that Redington's first two conditions for immunisation against small changes in the rate of interest are satisfied for this insurance company.

[8 marks]

(b) Determine the profit or loss, expressed as a present value, that the insurance company will make if the force of interest increases immediately to 5.5% per annum.

[5 marks]

(c) Explain how you might have anticipated, before making the calculation in part (b), whether the result would be a profit or loss.

[3 marks]

[Total 16 Marks]

5. (a) A random variable X is lognormally distributed with parameters μ and σ^2. The mean of X is 1.06 and its variance is 0.04.

The kth moment of X is

$$\mathbb{E}\left(X^k\right) = \exp\left(k\mu + \frac{1}{2}k^2\sigma^2\right), \quad \text{for } k = 1, 2, 3, ...$$

Determine the values of μ and σ.

[4 marks]

An investment fund provides annual rates of return, which are independent and identically distributed, with annual accumulation factors following the lognormal distribution with $\mu = 0.03001$ and $\sigma = 0.13536$.

An investor knows that she will have to make a payment of £8 000 in ten years' time.

(b) Calculate the amount of cash that she should invest now in order that she has a 97.5% chance of having sufficient cash available in ten years' time from the investment to meet the payment.

Use the fact that $\Phi(1.96) = 0.975$, where $\Phi(x)$ represents the value of the cumulative distribution function of the standard normal at $x \in \mathbb{R}$.

[7 marks]

(c) Explain the reason for the difference in values of the payment of £8 000, paid in ten years' time, and the amount of cash calculated in part (b).

[2 marks]

(d) Comment on whether investing in the fund, with the objective of meeting the future payment with a 97.5% chance, is a reasonable strategy.

[2 marks]

[Total 15 Marks]

6. Beginning with the definition of $\ddot{s}_{\overline{n}|}$, prove that

$$\ddot{s}^{(m)}_{\overline{n}|} = \frac{(1+i)^n - 1}{d^{(m)}},$$

for any positive integers n and m, where i is the effective rate of interest per unit time period.

[3 marks]

[Total 3 Marks]